

Business Process Automation

with Power Automate and Copilot Studio Agents

WSQ Course Code: TGS-2022017524 · 2-Day Hands-On Training · Modules 1–3

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W
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Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

CERTIFICATION

Microsoft Power Platform, AI and data analytics — 20+ years of training experience.

DELIVERS

WSQ courses on AI, business automation, data science and software engineering.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Let's Know Each Other

- Your name and organisation / role.
- Your experience with Power Automate, Copilot Studio or other automation tools (if any).
- One repetitive business process you would love to automate after this course.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

5

Be punctual; return from breaks on time.

6

75% attendance is required.

What You'll Learn

Automate repetitive business work with Microsoft Power Automate

Build AI-powered business agents in Microsoft Copilot Studio

Connect agents to flows for true end-to-end automation

Design real workflows for sales, finance, procurement and operations

- Practise in hands-on labs every day — you build working automations

Your 2-Day Journey

Day 1 — Foundations & Power Automate: automation logic, triggers, actions, 5 labs

Day 2 — Copilot Studio Agents: knowledge, tools, agent + flow, 6 labs + assessment

Each topic follows: concept → live demo → you build it

- You finish with working automations you can take back to work

How This Course Works

Concept, then demo, then hands-on — for every topic

Built entirely in Microsoft 365 and the Power Platform

No coding required — low-code, drag-and-drop tools

Save your flows and agents to reuse in your own role

- Daily recap and open Q&A to lock in what you learned

SCHEDULE

Lesson Plan — 2 Days, 9:00am–6:00pm

Day 1

- Day 1 — Foundations & Power Automate
 - Modules 1–2: workflow concepts, Power Automate
 - Labs 0–5: email, Excel logging, approval, scheduled, form flows

Day 2 & assessment

- Day 2 — Business Agents with Copilot Studio
 - Module 3: agents, prompts, agent + flow
 - Labs 6–11: first agent, knowledge (RAG), tools, agent flows
- Assessment (Day 2, 4:00–6:00pm)
 - Written Assessment 1 hr + Practical Performance 1 hr
- Daily timing
 - 9:00am–6:00pm · 1-hour lunch · tea breaks within

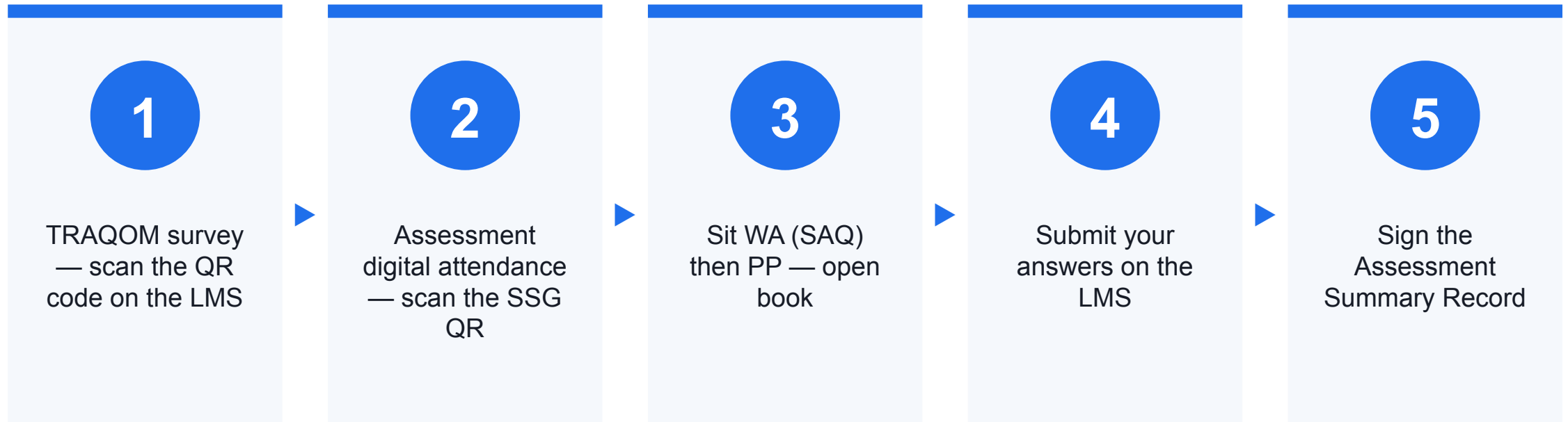
Briefing for Assessment

- Place phones and other materials under the table or on the floor.
- No photos or recording of assessment scripts.
- No discussion during the assessment.
- Use a black/blue pen for hard-copy assessments.
- No liquid paper / correction tape.
- Scripts are collected when time is up.

Assessment & Funding

- Written Assessment (SAQ) — 1 hour, open book. Tests the knowledge from Modules 1–3.
- Practical Performance (PP) — 1 hour, open book. You build a working flow + agent, as in the labs.
- Both are taken on Day 2, 4:00–6:00pm, and submitted on the LMS.
- WSQ funding requires 75% attendance AND passing the assessment (Competent).
- Courseware and the assessment are on the LMS: <https://lms-tms.tertiaryinfotech.com/>

Assessment Flow



Access the Hands-On Labs

All 12 labs (Labs 0–11) + the Learner Guide live in the course GitHub repository:

github.com/tertiarycourses/TGS-2022017524---Business-Process-Automation-with-Power-Automate-and-Copilot-Studio-Agents

Option A — Clone with Git

- Install Git (git-scm.com) if needed.
- `git clone <repo URL>.git`
- Open the labs/ folder in your editor or browser.

Option B — Download the ZIP

- Open the repo URL in your browser.
- Code → Download ZIP, then unzip.
- Open labs/ and follow each lab's index.md.

Day 1 — Foundations & Power Automate

Modules 1 & 2 · Understand automation, then build your first flows

Module 1 - Introduction to Workflow Automation

Why automate, and the building blocks behind every workflow

What Is Business Workflow Automation?

A workflow is a repeatable series of steps that gets work done

Automation lets software run those steps for you

It replaces manual, repetitive tasks — copying data, sending emails, chasing approvals

Runs reliably and consistently, with far fewer errors

- People focus on judgement; software handles the routine

Why Automate?

Save time — reclaim hours of manual work every week

Reduce errors — consistent, rule-based execution every time

Respond faster — instant notifications and approvals

Gain visibility — every step is logged and traceable

- Scale easily — handle more volume without more headcount

Manual Process vs. Automated Workflow

MANUAL PROCESS

Emails get missed in busy inboxes

Data re-keyed by hand between apps

Approvals stall and get forgotten

No audit trail of what happened

- Slow, inconsistent, hard to scale

AUTOMATED WORKFLOW

Triggers fire the moment an event happens

Data flows automatically between systems

Approvals routed, tracked and reminded

Every action logged automatically

- Fast, consistent and scalable

Standalone Agents vs. Integrated Workflows

STANDALONE AGENT

Answers questions and chats

Works inside the conversation

Output stays in the chat

- Great for information and guidance

INTEGRATED WORKFLOW

Takes action across systems

Connects apps, data and people

Produces results: records, emails, approvals

- Great for getting work done

Introduction to Workflow Logic

Every workflow follows the same simple pattern

Trigger → Action(s) → Output

Trigger: the event that starts the workflow

Actions: the steps the workflow performs

Output: the result the workflow produces

- Master this pattern and you can build any automation

Triggers — What Starts a Workflow

A trigger is the event that kicks the workflow off

Examples: an email arrives, a file is uploaded, a form is submitted, a schedule hits

Every flow has exactly one trigger — its starting gun

The trigger carries data into the flow: sender, subject, file name, form answers

- Choose the trigger that matches the business event you want to react to

Actions — What the Workflow Does

Actions are the steps performed after the trigger fires

Examples: send an email, add a row to Excel, post a notification, request an approval

They run in sequence — each step can reuse data from earlier steps

Combine many actions to model a complete business process

- Add conditions to decide which actions run

Outputs — The Result

The output is the outcome the workflow produces

A logged record, a sent email, an approval decision, a notification

Outputs can become the input to the next workflow

- Judge success by the output: did the right thing happen, reliably?

Workflow Steps — Putting It Together

Steps run top-to-bottom after the trigger

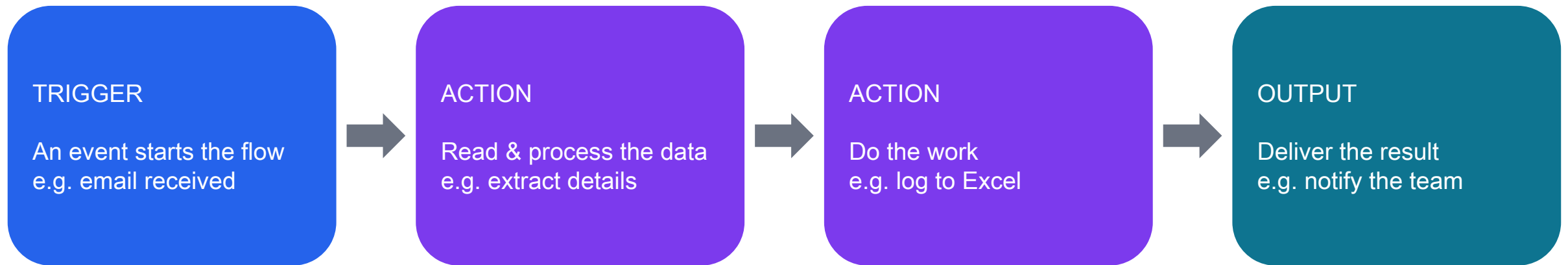
Add conditions (if / then) to branch the logic

Pass data between steps using dynamic content

Example: New email → read the details → log to Excel → notify the team

- Build complexity gradually — start simple, then extend

Anatomy of a Workflow



Master this pattern — Trigger → Action(s) → Output — and you can build any automation.

Module 1 Recap

Automation runs repeatable work for you — saving time and cutting errors

Standalone agents inform; integrated workflows take action

Every workflow follows: Trigger → Actions → Output

Triggers start it, actions do the work, outputs deliver the result

- Next: build real flows in Microsoft Power Automate

Module 2 - Introduction to Power Automate

Triggers, actions, and your first five working flows

Overview of Power Automate

Microsoft's low-code tool for automating tasks and processes

Connects 1,000+ apps and services — Outlook, Excel, SharePoint, Teams, Forms and more

Build flows visually: pick a trigger, add actions, done

Runs in the cloud — no servers, no code to maintain

- Part of Microsoft 365 and the Power Platform

Types of Flow

Automated — starts when an event happens (email arrives, file added)

Instant — you start it on demand with a button

Scheduled — runs on a timer (daily, hourly, weekly)

This course focuses on automated flows triggered by business events

- All are built in the same visual flow designer

The Maker Experience

Build flows at make.powerautomate.com

Designer canvas: trigger at the top, actions added below

Search a connector, choose an action, fill in the fields

Insert dynamic content to pass data between steps

- Test, then turn the flow on — it runs automatically from then on

Common Triggers

A trigger is the event that starts your flow

Email received — react to incoming mail in Outlook

File upload — a new file lands in SharePoint or OneDrive

Form submission — someone responds to a Microsoft Form

- Each trigger hands its data to the rest of the flow

Trigger: Email Received

Starts a flow when a new email arrives in Outlook

Filter by sender, subject, folder or importance

Grabs the email's data: from, subject, body, attachments

Use it for: enquiries, requests, alerts and notifications

- Lab example: a sales enquiry email kicks off logging and a reply

Trigger: File Upload

Starts a flow when a file is added to SharePoint or OneDrive

Watches a specific folder or document library

Captures the file name, content and metadata

Use it for: invoices, contracts, reports and submissions

- Lab example: an uploaded invoice starts an approval workflow

Trigger: Form Submission

Starts a flow when someone submits a Microsoft Form

Each answer becomes data you can use downstream

No inbox or folder to watch — the form is the front door

Use it for: requests, registrations, feedback and orders

- Lab example: a purchase request form routes to a manager for approval

Creating Workflow Actions

After the trigger, add the steps that do the work

Search for a connector, then pick the action you need

Fill in the fields — type text or insert dynamic content

Stack actions in order; reorder by dragging

- Three actions you'll use constantly: send email, add Excel row, request approval

Action: Send Emails

Sends an automated email from Outlook

Set the recipient, subject and body — any can be dynamic

Personalise with data from the trigger (sender name, request details)

Use for confirmations, notifications and auto-replies

- Keeps everyone informed without anyone lifting a finger

Action: Create Excel Entries

Adds a new row to a table in an Excel workbook

Map each column to data from the flow

Builds a running log automatically — enquiries, requests, transactions

Stored in OneDrive or SharePoint, ready to filter and report on

- Turns scattered events into a clean, structured record

Action: Notifications & Approvals

Notifications: post to Teams or send an alert when something happens

Approvals: send a request and wait for Approve or Reject

The approver responds by email, Teams or the Power Automate app

The flow pauses, captures the decision, then continues accordingly

- Perfect for sign-offs: invoices, purchases, leave and budgets

Connectors — Linking It All Together

A connector is the bridge to an app or service

Outlook, Excel, SharePoint, Teams, Forms, Approvals and 1,000+ more

Each connector offers its own triggers and actions

Sign in once; the flow uses your secure connection

- Mix connectors freely to span your whole toolset

Hands-On Labs - Power Automate

Build five working flows — email, Excel logging, approval, scheduled and form

Lab 1 — Automated Email Workflow

GOAL

Send a personalised email with one click

YOU'LL USE

Manual (instant) trigger + input

Dynamic content token

- Send an email action

STEPS

1. Trigger: Manually trigger a flow
2. Add a CustomerName input to the trigger
3. Action: Send an email (V2)
4. Personalise with the CustomerName token
- 5. Test → Manually → Run flow

Lab 2 — Excel Data Logging Workflow

GOAL

Log every submission to an Excel table

YOU'LL USE

Manual trigger with 3 inputs

Excel 'Add a row' action

- An Excel table in OneDrive

STEPS

1. Prepare an Excel table with columns
2. Trigger inputs: Name, Email, Message
3. Action: add a row to the table
4. Map the inputs + fx utcNow() date stamp
- 5. Test; optional: swap in a Forms trigger

Lab 3 — Simple Approval Workflow

GOAL

Route a request for Approve / Reject

YOU'LL USE

Manual trigger with 3 inputs

Start and wait for an approval

- Send an email action

STEPS

1. Trigger inputs: requester name, email, details
2. Action: start and wait for approval
3. Approver chooses Approve or Reject
4. Condition: branch on the decision
- 5. Email the outcome to the requester

Lab 4 — Scheduled Trigger Workflow

GOAL

Send a daily reminder / digest automatically

YOU'LL USE

Recurrence (schedule) trigger

Send an email action

- Optional: List rows from Excel

STEPS

1. Trigger: Recurrence — weekdays 9 AM
2. Set the correct time zone
3. Action: send a reminder email
4. Optional: add a live count from Excel
- 5. Test now, then let it run on schedule

Lab 5 — Form Submission Workflow

GOAL

Capture form submissions to email and Excel

YOU'LL USE

Microsoft Forms trigger

Get response details

- Send email + Add a row to Excel

STEPS

1. Build a Microsoft Form (shareable URL)
2. Trigger: when a response is submitted
3. Get response details
4. Email the team the new submission
- 5. Log the submission to an Excel table

Day 1 Recap

You learned what workflow automation is and why it matters

Every flow = Trigger → Actions → Output

Power Automate connects your apps with low-code flows

You built five working flows: email, Excel logging, approval, scheduled and form

- Tomorrow: build AI business agents in Copilot Studio

Day 2 — Building Business Agents with Copilot Studio

Module 3 · Create AI agents and connect them to your flows

Day 2 Roadmap

Recap: yesterday you automated tasks with Power Automate

Today: add intelligence with AI agents in Copilot Studio

Learn to design agents that produce clean, structured output

Connect those agents to the flows you already know

- Build six hands-on labs, then sit the WSQ assessment (4:00–6:00pm)

Module 3 - Building Business Agents with Copilot Studio

From a helpful chatbot to an agent that triggers real work

What Is Copilot Studio?

Microsoft's low-code tool for building custom AI agents

Create conversational agents that understand natural language

Ground them in your own knowledge — documents, sites, data

Agents can call Power Automate flows to take action

- Publish to Teams, websites and other channels

The Copilot Studio Maker

Build agents at copilotstudio.microsoft.com

Create a blank agent, then edit its details and instructions

Add knowledge sources, topics and tools

Test live in the built-in chat panel as you build

- Publish when you're happy — update any time

How an Agent Works

Instructions tell the agent its role and how to behave

Knowledge gives it the information to answer from

Topics handle specific conversations — the agent chooses one from its description

Tools let it do things — including running your flows

- It interprets the user, responds, and can hand off to automation

What Makes a Good Business Agent

A clear, narrow job — it does one thing well

A defined audience and tone of voice

Reliable, structured output the next step can use

Grounded in trustworthy knowledge — no guessing

- Knows when to act: hand off to a flow at the right moment

Creating Practical Business Agents

Start from a real task: 'handle sales enquiries', 'log procurement requests'

Write clear instructions: role, scope, what to collect, what to avoid

Add knowledge: product info, policies, price lists, FAQs

Decide the output: what should the agent hand back?

- Test with realistic questions before connecting any automation

Prompt Design Fundamentals

The prompt is the instruction that shapes the agent's response

Be specific: state the role, the task and the format you want

Give examples of good answers — the agent learns the pattern

Set boundaries: what to do when information is missing

- Iterate: test, refine the wording, test again

Prompt Design for Structured Outputs

A workflow needs predictable fields, not free-flowing prose

Ask the agent to return the same fields every time

Name each field clearly: Customer, Product, Quantity, Priority

Specify the format — a labelled list, a table, or JSON

- Structured output = data your Power Automate flow can read directly

Structured Output Formats

LABELLED FIELDS

Customer: Acme Ltd

Product: Widget Pro

Quantity: 50

Priority: High

- Easy to read, easy to map

JSON

```
{  
  "customer": "Acme Ltd",  
  "product": "Widget Pro",  
  "quantity": 50,  
  "priority": "High"  
}
```

- }

Example: A Structured Prompt

THE PROMPT

“You are a sales enquiry assistant.

From the customer's message, extract:

Customer name, Product, Quantity,
and Priority (High / Medium / Low).

Return the result as labelled fields.

- If a detail is missing, write ‘Not stated’.”

THE OUTPUT

Customer: Acme Ltd

Product: Widget Pro

Quantity: 50

Priority: High

- → Ready to log to Excel and notify sales

Connecting Agents to Power Automate Flows

An agent can call a flow as one of its actions

Build the flow with an 'instant' trigger that accepts inputs

Add the flow to your agent in Copilot Studio

The agent passes data in; the flow runs and returns a result

- This is how a conversation turns into real action

Passing Outputs into Workflows

The agent's structured output becomes the flow's input

Each field maps to a flow parameter: Customer, Product, Quantity

The flow uses those values in its actions — log, email, approve

Clean structured output makes mapping effortless

- Garbage in, garbage out — good prompts pay off here

Input and Output Mapping

Define the flow's inputs to match the agent's fields

Map each agent field to the matching flow parameter

The flow can return a value back to the agent (e.g. a reference number)

The agent confirms to the user: 'Logged — your reference is 1042'

- Data flows both ways: agent → flow → agent

Best Practices for Agent + Flow

Keep each flow focused on one clear job

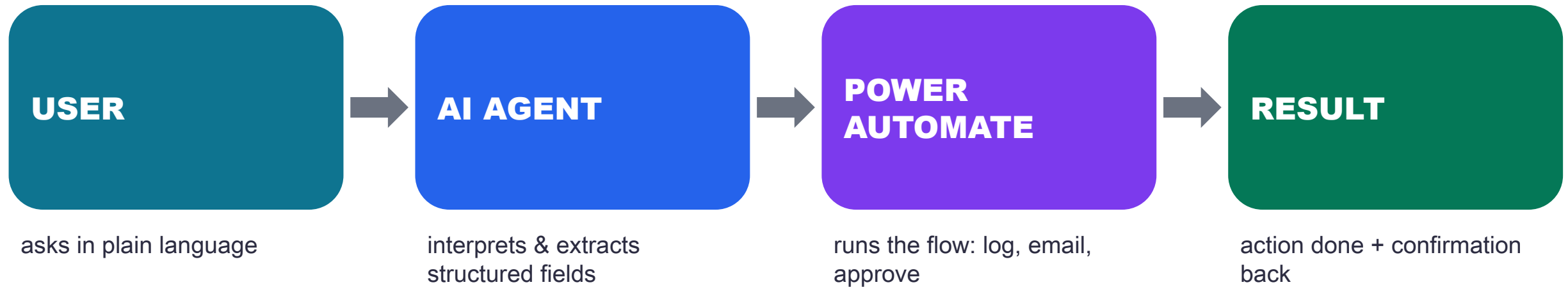
Validate inputs — handle missing or odd data gracefully

Use clear field names shared between agent and flow

Test the flow on its own before wiring it to the agent

- Tell the user what happened — confirm success or explain failure

How Agents and Power Automate Work Together



The agent handles the conversation and the thinking; Power Automate does the work across your systems.

Hands-On Labs - Copilot Studio

Six labs: create an agent, add knowledge & tools, then three business agents

Lab 6 — Create Your First Agent

GOAL

Build, configure and test your first Copilot Studio agent

YOU'LL USE

Copilot Studio agent

- Instructions + built-in Test pane

STEPS

1. Create a blank agent (Agents → Create blank agent)
2. Write clear instructions (role + tone)
3. Add a knowledge source
4. Test it live in the chat panel
- 5. Refine instructions, then publish (needs a full licence)

Lab 7 — Add Knowledge to Your Agent (RAG)

GOAL

Ground the agent in your own documents (RAG) with citations

YOU'LL USE

Knowledge sources (files / website)

- Copilot Studio agent

STEPS

1. Prepare a document or website
2. Add it as a knowledge source
3. Wait until the source is Ready
4. Ask questions answered from your content
- 5. Check the citations; tune the sources

Lab 8 — Add Tools and Actions

GOAL

Give the agent tools so it can act, not just answer

YOU'LL USE

Connector / prebuilt action

- A Power Automate flow as a tool

STEPS

1. Open the Tools tab, add a tool
2. Add a connector action (e.g. send email)
3. Describe when the agent should use it
4. Or add a flow as a tool with inputs
- 5. Test the agent performing the action

Lab 9 — Sales Enquiry Assistant

GOAL

An agent that captures sales enquiries as structured fields

YOU'LL USE

Copilot Studio agent

- A topic with Ask-a-question nodes

STEPS

1. Create the Sales Enquiry Assistant agent
2. Add a "New Sales Enquiry" topic — the agent chooses it from its description
3. Ask-a-question nodes save Name, Company, Product, Quantity
4. Show a structured summary of all four variables
- 5. Test — Quantity uses Identify = Number, so text re-asks

Lab 10 — Procurement Request Workflow

GOAL

An agent that takes procurement requests and starts an approval flow

YOU'LL USE

Copilot Studio agent

- A Power Automate flow (approval)

STEPS

1. Build an agent to collect request details
2. Create a flow with an approval step
3. Add the flow as a tool in the topic
4. Pass the request fields into the flow
- 5. Test end-to-end: request → approval

Lab 11 — Automated Response Generation

GOAL

An agent that drafts a tailored reply from the captured details

YOU'LL USE

Agent flow + AI Builder Run a prompt

- Send an email (V2) action

STEPS

1. Reuse the enquiry details from Lab 9
2. Agent flow runs an AI Builder 'Run a prompt'
3. Keep the tone professional and on-brand
4. Email the draft, then Respond to the agent
- 5. Review, refine the prompt, repeat

Testing & Debugging Agents

Test in the chat panel with realistic, messy inputs

Check the output format is consistent every time

Trace flow runs to see exactly what data passed through

Refine the prompt when answers drift or fields go missing

- Handle the edge cases: missing info, unexpected questions

Day 2 Recap

Copilot Studio builds low-code AI business agents

Good prompts produce structured, predictable output

Agents connect to Power Automate to take real action

You created an agent, added knowledge and tools, then built a sales assistant, a procurement agent and a response generator

- Next: course wrap-up, then the WSQ assessment (4:00–6:00pm)

What You Can Automate Next

Onboarding: new-hire checklists and welcome emails

Reporting: scheduled data pulls and summaries

Helpdesk: triage and route incoming requests

Document handling: filing, approvals and reminders

- Anywhere people copy data or chase approvals by hand

Common Pitfalls to Avoid

Automating a broken process — fix the process first

Trying to build everything in one giant flow

Skipping tests, then surprises in production

No error handling — failures go unnoticed

- Vague agent prompts that return inconsistent data

Common Errors & Quick Fixes

“Unauthorized” sending email — reconnect Outlook with a mailbox-enabled account

Approval ‘valid users’ error — pick the approver from the dropdown (a real user in your tenant)

Date shows literal utcNow() — enter it via the fx / Expression editor so it becomes a token

Excel cell shows ##### — the column is just too narrow; auto-fit it

- Unwanted ‘For each’, or agent can’t see its flow — use single-value fields; same environment in both tools

Governance & Security Basics

Use connections and permissions you're authorised to use

Keep sensitive data out of emails and logs where possible

Follow your organisation's data and approval policies

Document what each flow does and who owns it

- Review and retire flows that are no longer needed

Rolling This Out at Work

Pick one painful, repetitive task to automate first

Prove the value, then share the win with your team

Build a small library of reusable flows and agents

Involve the people who do the work in the design

- Iterate — improve flows as needs change

Course Recap — The Big Picture

Day 1: automation logic and Power Automate flows

Day 2: business agents in Copilot Studio, wired to your flows end to end

You built working automations for real scenarios

- You can now design, build and run your own workflows

Key Takeaways

Every workflow is Trigger → Actions → Output

Power Automate does the work; Copilot Studio adds the intelligence

Structured agent outputs make flows reliable

Orchestration turns steps into complete processes

- Start small, test, and improve — then scale

Resources & Where to Go Next

make.powerautomate.com — build and manage your flows

copilotstudio.microsoft.com — create and publish agents

Microsoft Learn — free guided learning paths

Templates gallery — start from pre-built flows

- Your community — share flows and patterns with colleagues

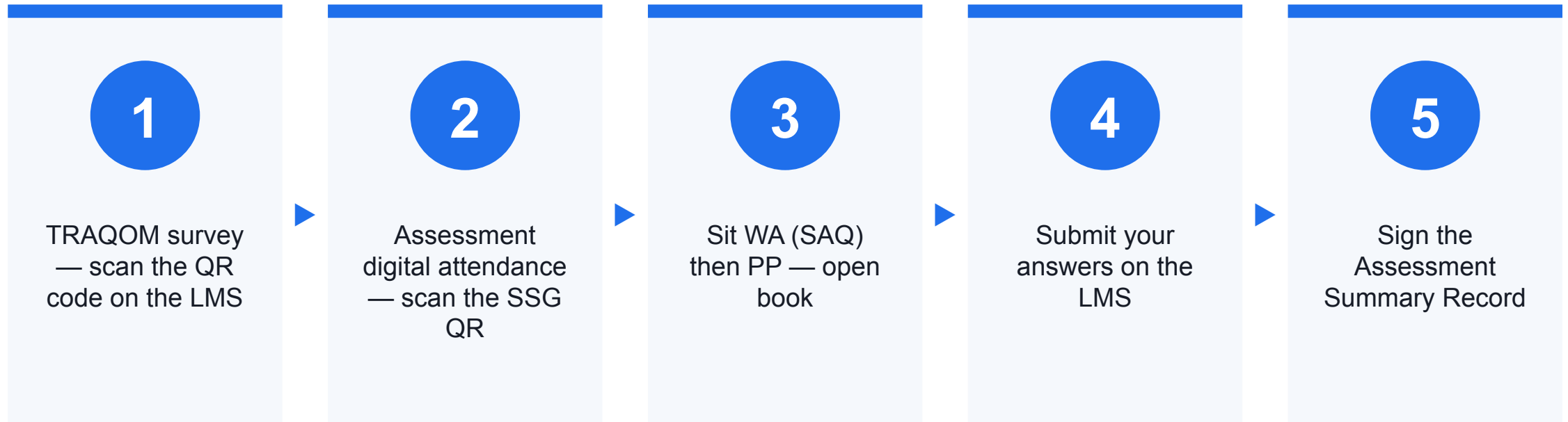
Courseware & Assessment on the LMS

- Download the slides and the Learner Guide from the LMS for the open-book assessment.
- Portal: <https://lms-tms.tertiaryinfotech.com/>
- Submit your Written Assessment and Practical Performance answers on the LMS.
- Complete the TRAQOM survey via the QR code on the LMS.

Assessment

- Written Assessment (SAQ) — 1 hour. Practical Performance (PP) — 1 hour.
- Open book: slides, Learner Guide and approved materials only.
- Remember to take the Assessment digital attendance (TRAQOM).
- Submit your completed answers on the LMS at <https://lms-tms.tertiaryinfotech.com/>.

Assessment Flow



Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Thank You

Now go automate something — questions welcome